

Case Study 4: Taxi Demand Prediction in NYC

V1: Business/Real World Problem Overview

V2: Business objective and Constraints

End user: Taxi driver

Obj: Predict the # of pickups as accurately as possible for each region in a 10 min interval.

Break time in 10 min interval and region in NYC.

Constraints: Latency

Interpretability

Relative errors $\rightarrow$ percentage error

V3: Mapping to ML problem: Data

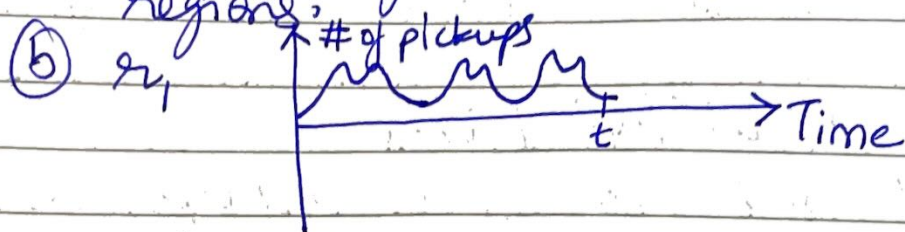
V4: Mapping to ML problem: data data frames

dask optimizes large datasets that can be used in smaller RAMs

V6: Time Series, forecasting/Regression

① region, 10 min interval predict # of pickups

② How do you breakup NYC into regions.



from raw data, I have, longitude, latitude and pickup.

Predict for time $t+1$, the no. of pickups.

$$P_{t+1} = ?$$

$$P_{t+1} = P_t + \alpha$$

Time Series prediction

V7: Performance Metrics

- 1) Mean Absolute Percentage error.
- 2) Mean Squared error.

$$1) \frac{100}{n} \sum_{i=1}^n \frac{|P_t - \hat{P}_t|}{P_t}$$

$$2) \frac{1}{n} \sum_{i=1}^n (P_t - \hat{P}_t)^2$$

V8: Data cleaning: Latitude and Longitude data

V9: DC: Trip duration

V10: DC: Speed

V11: DC: Distance

V12: DC: Fare

V13: DC: Removing all outliers/erroneous points

Estimate how much data you are losing after data cleaning. Try on some chunk of data and then for whole data.

V14: Data Preparation: Clustering/Segmentation

V15: DP: Time binning

V16: DP: Smoothing time-series data

V17: Contd.

V18: DP: Time Series and Fourier Transforms

Study Fourier Transforms, esp. DC component.

V19: Ratio and previous-time-bin values

cleaned outliers and error.
prepared clusters using k-means
and 10-min bin.

Features:-

1) Ratios

eg - t_{16} = Jan 2016

t_{15} = Jan 2015

$$R_t = \frac{p_t^{2016}}{p_t^{2015}}$$

$$R_{t+1} = \frac{p_{t+1}^{2016}}{p_{t+1}^{2015}} \Rightarrow$$

V20: Simple moving average

$$P_t^{2016} = P_t^{2015} * R_t$$

R_t is calculated using simple moving average.

$$R_t = \frac{(R_{t-1} + R_{t-2} + R_{t-3} + \dots + R_{t-n})}{n}$$

Take most recent n values.

MAPE - Mean absolute Percentage Error

$$\frac{1}{n} \sum_{t=1}^n \frac{P_t^{2016} - \hat{P}_t^{2016}}{P_t^{2016}} * 100$$

Window Size = 3. (n which is HP)

V2.1: Weighted Moving average

Most recent values of time have higher weightage.

$$P_t = \frac{n * P_{t-1} + (n-1)P_{t-2} + \dots + 1P_{t-n}}{\frac{n(n+1)}{2}}$$

V2.2: Exponential Weighted Moving Average

$$R_t' = \alpha R_{t-1} + (1-\alpha)R_{t-1}'$$

| | |
predicted observed value predicted value

$0 < \alpha \leq 1$

V2.3: Results

V2.4: Regression Models: Train-Test split & Features

V2.5: Linear Regression

V2.6: Random Forest Regression

V2.7: Xgboost Regressor

28: Model Performance

Assignment 16

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Storage bucket → m-16/2016-01-Train
2016-02-Test

NYC-FINAL → All important imports done.
Data uploaded - In progress.